ChainSight — AI-Powered Supply Chain Visibility Platform

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Target: Malaysia (Penang) | Semiconductor Supply Chain

Value Statement **CHT leverages AI-native 5G-Advanced, IoT, and its Malaysian subsidiary** to deliver a cross-border smart factory visibility platform — enabling Penang's 3,000+ semiconductor SME suppliers to share real-time production data with Tier-1 clients, **reducing supply chain response from 48 hours to instant**, while securing OSAT supply chains against geopolitical risks with 24/7 resilience.

PROBLEM & OPPORTUNITY

"When Intel asks a Penang supplier 'what's your real capacity?', the answer takes 48 hours — by then, the order has gone to Vietnam."

- **Geopolitical shift:** China+1 & Taiwan+1 accelerate OSAT relocation → demand for "Just-in-Case" supply chains
- **Visibility gap:** **3,000+** Tier 2-3 SMEs in Penang still use Excel + phone. Only **18%** adopted Industry 4.0
- **Investment boom:** Intel **$7B** + Infineon €5B (Kulim) + ASE 5th plant
- **Policy window:** NIA framework + NIMP 2030 (3,000 smart factories). CHT MY subsidiary est. Dec 2025

AI-ENABLED SOLUTION

Lightweight factory digitalization for semiconductor SME suppliers — 3 AI modules:

Pulse	Predict	Comply
5G IoT → real-time OEE, yield on client Portal. AI auto-classifies machine status.	ML predicts failure 7-14 days ahead, material depletion, true delivery dates.	Auto-maps to RBA/CSRD/OSH Act. One-click audit reports.

Why AI is essential: Transforms raw IoT signals into actionable predictions — decisions that required engineers in short supply (MY needs 60K; produces 5K/yr).

INNOVATION

- **Position:** Not another ERP — fills the real-time gap between ERP (lagging) and factory floor (live) via 5G IoT. **Complements SAP, not competes.**
- **Biz Model: "Fabs pay, SMEs deploy free."** Intel/ASE pay Portal subscription → subsidize SME IoT install → triple-win.
- **Network Effect:** More suppliers → more visibility → fabs pay more → **flywheel** that competitors without telecom infra cannot replicate.
- **Cross-border:** Taiwan HQ monitors MY factories via CHT submarine cables + SD-WAN — 24/7 resilient.

TARGET CUSTOMERS & MARKET

- **Side A — Fabs:** Intel, ASE, Infineon, Inari → Portal **$5-15K/mo** for supplier visibility + compliance
- **Side B — SMEs:** 3,000+ Penang SMEs → zero upfront + **$800-2K/mo** for AI + compliance
- **Taiwanese firms:** secure cross-border factory control under Taiwan+1 relocation

TAM	SAM	SOM Yr3
$41-65B	$1.2B	$6.6M

Global supply chain AI 2030 → SEA semiconductor → 8 fabs + 120 SMEs

EXPECTED IMPACT

Stakeholder	Key Metrics	Before	After ChainSight
Fabs	Response / Disruption warning	48hr / After-the-fact	Real-time / 7-14 days ahead
SMEs	Downtime / Inventory / Compliance	High / Cash tied up / 3ppl×2mo	-50% / 30-40% freed / 1 click
CHT	Year 3 revenue	—	$4.2M ARR (64% recurring)

Taiwanese Ecosystem
Digital umbilical cord — TW HQ monitors southbound factories real-time. Semicon first → PCB, machinery next.

Geopolitical Resilience
Non-China, non-US data trust layer — the missing software infra after China+1 relocation.

Policy & ESG
TW New Southbound × MY NIMP 2030 flagship demo. Comply enables whole-chain CSRD compliance.

FEASIBILITY & SCALABILITY

Why only CHT? 5G private network + iEN IoT platform + 14+ submarine cables + cybersecurity subsidiary + MY subsidiary (Dec 2025). Proven tech, no invention needed. **CHT does the infra layer SAP cannot; domain AI co-built with semiconductor partners.**

$3M Budget — 3 Phases:

Phase 1 (M1-6) $900K
Pulse + 3 SME pilots + 1 fab Portal
KPI: 3 live dashboards, <2s latency

Phase 2 (M7-12) $1.2M
Predict + 30 SMEs + 3 fabs
KPI: 85% prediction accuracy, $1.2M ARR

Phase 3 (M13-18) $900K
Comply + 80 SMEs + full Penang
KPI: 80 SMEs onboarded, $4.2M ARR

Scale Path
Penang → Kedah → Johor → Vietnam → Thailand → **all SEA**. Each country replicates the same fab-subsidized playbook with local CHT/partner subsidiary.

Risk Mitigation
SME adoption: fab-subsidized, zero upfront
Data security: edge compute + E2E encryption
ERP overlap: supplement SAP, not replace